

GENERAL SPECIFICATION G2.22

# Project Quality and Execution

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## G2.22 PROJECT QUALITY AND EXECUTION

### G2.22.1 Project Quality and Assurance Plan

#### (a) General

The Contractor's Project Quality and Assurance Plan shall comply with ISO 9000 or its equivalent. Where the Contractor is a Joint Venture firm or Partnership, the Project Quality and Assurance Plan shall follow that of the leading partner or partner with the ISO certification and the Project shall be treated as that partner's Project Quality and Assurance Plan for ISO auditing purposes. Internal quality audit and external quality audit shall be carried out at 6-monthly and yearly intervals respectively. A copy of all Quality Audit (internal and external) results, reports and corrective actions shall be submitted to the S.O..

#### (b) Project Quality Plan

(i) Within four (4) weeks of award of the Contract, the Contractor shall submit a preliminary Project Quality and Assurance Plan (PQP). The preliminary PQP shall indicate the approach and structure that the detailed plan will take and shall include the following:

- 1) A summary of the Contract requirements including all proposed quality adjectives;
- 2) A list of the quality assurance and quality control procedures proposed by the Contractor for his use in the execution of the Works;
- 3) A list of all the Codes of Practice, Standards and Specifications that the Contractor proposes to apply to the work;
- 4) The Contractor's proposals for internal and external and sub-contractor's quality assurance audits;
- 5) A statement detailing the quality records that the Contractor proposes to keep, the time during which they will be prepared and the subsequent period and manner in which they will be stored;
- 6) Inspection and testing plans for every activity requiring inspection. The plans shall identify the level of inspection required and who is responsible for releasing an activity from a "Hold Point";
- 7) Standard templates to be used for the implementation of the PQP as approved by the S.O.; and
- 8) For Operational Phases of the Project (as required), the PQP shall be updated to include the requirements of G2.22 for the operational activities. The PQP shall be submitted for the S.O.'s approval, 3 months prior to the start of the Operational Phase.

(ii) Within twelve (12) weeks of award of Contract, the Contractor shall provide the S.O., with a detailed Quality Plan taking into account any directions or requirements from the S.O. on the preliminary plan.

(iii) The detailed Quality Plan shall be updated, expanded and improved as necessary from time to time, but not less than annually, to incorporate, for the S.O.'s acceptance, all changes to the Contractor's procedures.

(c) Organisation Chart

The Contractor shall submit a detailed organization chart identifying the responsibilities, authority and interrelation of all personnel who manage, perform and verify work affecting quality. The organization chart shall be specific only to this Contract. The chart shall identify the Quality Management Representative who shall act as the Quality Coordinator for the Contractor in all dealings with the S.O. for approval by the S.O..

(d) Control of Testing and Measuring Instruments

All testing and measuring information shall be tested or calibrated at regular stated periods by approved Authorities with traceability to a National Standard. Copies of test/calibration results, reports etc. shall be submitted to the S.O.. These records shall also form part of the Quality Records.

(e) Identification and Traceability

The Contractor shall produce and maintain procedures for identifying products from applicable drawings, specifications and other documents during all stages of production, delivery and installation. Traceability of materials and equipment shall be documented and managed. Notwithstanding the requirements of the Contractor's quality system he shall retain all inspection, test, conformity and release for inspection by the S.O. when requested.

(f) Quality Assurance Inspection and Testing

- (i) The Contractor shall implement all necessary Quality Assurance / Quality Control procedures for the Works as specified in the Specifications by which the Contractor ensures that the Works, including that performed by his sub-contractors and suppliers, complies with the requirements of the Contract.
- (ii) All Works undertaken in this Contract shall be subject to the quality assurance inspection and testing at all locations and at all reasonable times before acceptance to ensure strict compliance with the terms of the Contract Documents.
- (iii) These inspections and tests do not relieve the Contractor of responsibility for providing adequate quality control measures or relieve the Contractor of the responsibility for damage to or loss of the material before acceptance. The tests and inspections do not constitute acceptance or affect the continuing rights of the Board after acceptance of the completed work.
- (iv) Standard templates issued by S.O. to be used for the Inspection and Test Plans, Reports and Checklists.

(g) Quality Audit

The Contractor shall make available on request any documents, which relate to his recent internal and external audits including other quality records. The S.O. may also conduct compliance audits of the Contractor's quality system for which the S.O. shall give not less than two (2) weeks' notice. During any audit by the S.O., the Contractor shall provide suitably qualified staff to accompany the auditor.

- (h) The contractor shall define the policy for quality, including objectives for quality and its commitment to quality specifically relevant to the Contract.

- (i) Safety Statement  
The Contractor shall define a Safety Statement and make reference to other safety documents pertaining to the Contract.
- (j) Design Control  
The Contractor shall establish and maintain documented procedures to control and verify the design in order to ensure that the specified requirements in the Contract are met. The control shall provide for design and development planning, organizational and technical interfaces, design input, design output, design review, design verification, design validation and design changes. This process to be included in the PQP to the S.O.'s approval.
- (k) Document and Data Control  
The Contractor shall establish and maintain documented procedures to control all documents and data that relate to quality including standards and drawings from the S.O. and other regulatory bodies.
- (l) Purchasing  
The Contractor shall establish and maintain documented procedures to ensure that purchased products conform to specified requirements. The control shall provide for evaluation of sub-contractors, control of purchasing data and verification of purchased products.
- (m) Control of Non-Conforming Product  
The Contractor shall establish and maintain documented procedures to ensure that product that does not conform to specified requirements is prevented from unintended use or installation. The control shall provide for identification, documentation, evaluation, segregation, disposal of non-conforming product and notification of functions concerned. The S.O. to be informed of all Non-Conforming Products.
- (n) Corrective and Preventive Action (CAPA)  
The Contractor shall establish and maintain documented procedures for implementing corrective and preventive action to eliminate the causes of actual or potential non-conformities. The Contractor shall implement and record any changes to the documented procedures resulting from corrective and preventive action. The S.O. to be informed of all CAPAs.

#### **G2.22.2 Information for the Execution of Works**

- (a) The Contractor is advised that it shall be deemed to be his sole responsibility to ascertain for himself the extent of work that is required to be done on site and to generally obtain his own information on all matters affecting the execution of the whole of the works involved in this Contract to the entire satisfaction of the S.O.. No claim for extras in consequence of any alleged ignorance in any aspect will be entertained by the Government. It must be clearly and definitely understood that the Contractor shall be held solely responsible for making all necessary arrangements and co-coordinating with all relevant Authorities, Specialist sub-contractors, etc. to ensure satisfactory completion of this Contract.

- (b) Although various buildings, sewers, drains, etc. are shown on the drawings, the Contractor will be held solely responsible for making his own investigations and for allowing for any extra costs in his sum for all obstructions, whether indicated in the drawings or not.

#### **G2.22.3 Protection Against Damage**

- (a) The Contractor shall take all necessary precautions to avoid causing any unwarranted damage to roads, land, properties, trees and other features and, during the course of the Contract, shall deal promptly with any complaints by the Board.
- (b) Where any portion of the Works is close to, across, or under any existing apparatus of Public Utilities or other parties, the Contractor shall work around, under or adjacent to all services in a manner designed to avoid damage, leakage or danger, and to ensure uninterrupted-operation.
- (c) Should any leakage or damage be discovered, the Contractor shall at once notify the S.O. and the Statutory Authority concerned, and the Contractor shall afford every facility for the repair or replacement of the apparatus affected.
- (d) The Contractor shall be responsible for the preservation of all public and private properties and shall use every precaution necessary to prevent damage thereto. If any direct or indirect damage is done to public or private properties by or on account of any act, omission, neglect or misconduct in the execution of the work on the part of the Contractor, such property shall be restored by the Contractor at his own expense to a condition similar or equal to that existing before the damage was done or he shall make good the damage in other manner acceptable to the S.O..
- (e) All repair works shall be completed within reasonable periods to the satisfaction of the S.O.. The Contractor shall submit his proposal for the remedial works to the S.O. or approval prior to commencement of such works. In addition, the Contractor shall bear all direct and consequential damages and losses arising from the execution of the works.
- (f) The Contractor is responsible for restricting his workman to the site of the work. No equipment and material shall be stored at places other than those areas allocated to him by the S.O.. Temporary barricading shall be erected to separate the worksite from the rest of private premises at the contractor's own expense.
- (g) The Contractor shall include in his sums for complying with all the foregoing requirements of this clause.

**END OF SECTION**